



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

FIRST SEMESTER 2026-2027

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the time-table) this portion gives further specific details regarding the course.

Course No : **EEE F314**
Course Title : **FPGA BASED SYSTEM DESIGN**
Instructor-in-Charge : **GOVIND PRASAD** (govind.prasad@pilani.bits-pilani.ac.in)
Lecture: Govind Prasad
Practical: Divya Vyas (p20210431@pilani.bits-pilani.ac.in)
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1. Course Description:

Digital system design options and trade-offs, design methodology and technology overview, Programmable Logic Devices, SPLDs, CPLDs, and Field Programmable Gate Arrays (FPGAs); FPGA architectural options, granularity of function and wiring resources, coarse vs. fine-grained, vendor-specific issues (emphasis on Xilinx and Altera), Logic fabric, Look-up tables, block BRAMs, DSP slices etc., timing models, power dissipation, Input and Output cell characteristics, clock input, Timing, Power dissipation, Programmable interconnect, Hardware Description Languages, FPGA system design tools, synthesis, and implementation, design verification and debugging, FPGA based design studies, interfacing with input-output devices and data converters.

2. Scope and Objective of the Course:

This course aims to upskill the students with the following topics:

- Build a strong foundation in FPGA architecture, technologies, and design methodologies.
- Develop proficiency in HDL-based design for creating complex digital systems.
- Explore the full FPGA design flow—from concept to implementation and debugging.
- Integrate embedded processors, IP cores, and DSP functions into FPGA-based designs.
- Apply knowledge through hands-on labs and projects using modern FPGA platforms.

3. Course Learning Outcomes: Students will gain a comprehensive understanding of FPGA architecture and design flow, develop proficiency in HDL-based digital design, and learn to integrate digital system, embedded processors, and IP cores. They will also acquire hands-on experience with modern FPGA tools and platforms to implement, debug, and optimize complex digital systems.

4. Text Book:



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TB1: Wolf Wayne, FPGA Based System Design, Pearson Edu, 2004.

Reference Books:

RB1: Samir Palnitkar, Verilog HDL: A Guide to Digital Design and Synthesis, Pearson Education, Asia, 2003.

RB2: Bora Tar and Cem Ünsalan, Digital System Design with FPGA: Implementation Using Verilog and VHDL, McGraw-Hill, 2017.

RB3: J. Bhasker, Verilog HDL Synthesis: A Practical Primer, Star Galaxy.

RB4: FPGA data sheet and user guide

RB5: User guide of the design tool

5. Method of Conduct of the Course: Offline including Blackboard and PPT

6. Course Plan:

Lecture No.	Learning Objectives	Topics to be Covered	Ref.
01	Introduction	Digital System design options and trade-offs	TB1
02	Programmable Logic Devices	SPLDs, CPLDs and Field Programmable Gate Arrays (FPGAs)	TB1
03-04	FPGA Architecture	Coarse vs fine-grained architectures, SRAM Based-FPGAs, Permanently Programmable FPGAs, Programmable I/O, Look-up tables, block BRAMs, DSP slices etc. SoC FPGAs,	TB1, RB1, RB2, RB3
05-06	Vendor Specific Architectures	Case Studies (Xilinx, Altera, Microsemi, etc.)	TB1, RB3
07-08	FPGA Design Flow	FPGA-based system design methodology - Simulation, synthesis and design implementation - Debugging and verification	TB1, RB4, RB5
09-10	Performance Analysis	Area, speed and power-aware design, Timing models, power analysis and clocking	TB1, RB4
11-12	FPGA-Based Design Examples	Signal or image processing applications, interfacing with input-output devices and data converters.	TB1, RB1

7. List of Experiments:

Lab experiments covering the following topics:

- FPGA Design Flow
 - Combination circuits
 - Sequential circuits
 - Mealy and Moore state machines



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- IP Cores
- Debugging and verification using VIO and ILA
- Interfacing IOs
- Mapping signal processing/image processing algorithms in FPGA
- ADC/DAC interfacing for real time applications
- SPI and I2C protocol implementation
- Scalable FSM-based memory management scheme (e.g., ping-pong, FIFO)
- FPGA-based design projects

8. Evaluation Scheme:

SL. No.	Component	Duration	Marks (Weightage)	Date & Time	Nature of component
1.	Regular Lab Performance	Continuous	10 (10%)	Regular and announced	OB
2.	Lab Assignments	Continuous	15 (15%)	To be announced	OB
3.	Mid-Sem Exam	90 min	25 (25%)	9/10/2026	CB
4.	Lab Project	Continuous	25 (25%)	To be announced	OB
5.	Comp. Lab Exam	3 hours	25 (25%)	To be announced	OB+CB

9. Assignments /Project: Regular assignments and project work involving the use of FPGA boards with Verilog for simulation and physical verification of advanced VLSI circuits.

10. Regular Lab Performance: Evaluation based on regular lab performance, including viva conducted during lab sessions.

11. Make up Policy: Make up will be given only on genuine reasons. Applications for make up should be given in advance and prior permission should be obtained for Scheduled tests.

12. Chamber Consultation Hours: Friday 3-4PM.

13. Notices: All notices related to the course will be put on NALANDA.

Instructor-In-Charge
EEE F348



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